

# Yuzhen Song

Southern University of Science and Technology

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Personal Page • GitHub Repository

**Research Interests:** Humanoid Locomotion and Whole-Body Control; Reinforcement Learning; Sim-to-Real Transfer.

## EDUCATION

**Southern University of Science and Technology**

B.Eng. in Robotics Engineering

**Shenzhen, China**

Aug 2023 – Jun 2027

- **Overall GPA:** 3.87/4.0 (Weighted Avg: 93.03/100)    **Major GPA:** 3.94/4.0
- **Core Courses:** Robot Operating System (100), Mechanics of Materials M (100), Mechatronic Systems (98), Robot Modeling and Control (97), Mechanical Design (97), Finite Element Theory (97), Intro to C Programming (97), Signals and Systems (95), Sensing Technology (95), Linear Algebra (94), Machine Learning for Eng. (93), Robotic Actuation System (93), Computer Vision (90), Fundamentals of Control Eng. (90)
- **Leadership:** President of Robotics Club & Student Organizations Council (SOC)

**Visiting Student**

University of California, Berkeley

Summer Sessions (Session E); Berkeley Global Access (BGA) Program (Fall)

**Berkeley, CA, USA**

Jul 2026 – Dec 2026

## PUBLICATIONS

- J. Cao\*, Y. Chen\*, **Y. Song**, M. Tomizuka, C. Li, T. Tian, “**TEXEDO: Test Time Scaling for Controller-Aware Language-Conditioned Humanoid Motion Generation**,” *Conference on Robot Learning (CoRL)*, 2026. **Accepted**.  
Also presented at the WCBM and RoboData workshops at RSS 2026.

## RESEARCH EXPERIENCES

**University of California, Berkeley**

Visiting Undergraduate Researcher, Advisor: Prof. Masayoshi Tomizuka

**Berkeley, CA, USA**

Apr 2026 – Sept 2026

- **Project: TEXEDO – Test-Time Scaling for Controller-Aware Humanoid Motion Generation**
  - ▷ Integrated Kimodo motions into MuJoCo via mjlab, aligning quaternion conventions, sampling rates, and evaluation logs.
  - ▷ Compared TEXEDO with Kimodo using MPJPE, velocity errors, and acceleration errors; resolved memory failures through streaming computation and logging.
  - ▷ Built a Gemini video evaluator for semantic alignment and physical plausibility, comparing automated scores with manual ratings.
  - ▷ Co-authored TEXEDO, accepted to **CoRL 2026** and presented at the RSS 2026 WCBM and RoboData workshops.

**Southern University of Science and Technology**

Undergraduate Researcher, Advisor: Prof. Chenglong Fu

**Shenzhen, China**

Sep 2024 – Present

- **Project: A Model for Predicting Metabolically Optimal Anteroposterior Center-of-Mass Assistance During Uphill Walking**
  - ▷ Reproduced the team’s equations for single and double support and implemented code to predict and plot optimal assistance across slopes.
  - ▷ Designed a cable assistance platform with a motorized spool, a rail and pulley mechanism for angle adjustment, and a waist belt attachment.
  - ▷ Participating in ongoing treadmill experiments to evaluate metabolic responses to predicted assistance.

Undergraduate Researcher, Advisors: Prof. Jiansheng Dai, Prof. Zhao Tang

Jun 2025 – Oct 2025

- **Project: Simulation of an Underactuated Metamorphic Palm for a Three-Fingered Hand**
  - ▷ Built Isaac Sim and Webots environments for an underactuated palm with five links, adding closure constraints to URDF models for preliminary RL experiments.
  - ▷ Simplified meshes while preserving geometry at contact regions, resolving GPU memory failures and enabling simulation on an 8GB GPU.
  - ▷ Investigated instability during palm reconfiguration by analyzing joint constraints and tuning damping, stiffness, and friction, identifying limitations in simulation fidelity.

## SELECTED PROJECTS

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### Southern University of Science and Technology

Project Leader, Advisor: Prof. Chengzhi Hu

Shenzhen, China

Mar 2026 – Jun 2026

- **Project: BEAT – Barometric Elevation and Attitude Tracking for Posture-Independent Wrist Blood Pressure Measurement**
  - ▷ Led BEAT development, using barometric sensing to correct for wrist height relative to the heart.
  - ▷ Programmed two ESP32 boards for ESP-NOW communication and built a desktop UI for live sensor monitoring.
  - ▷ Designed and analyzed pilot tests across three arm positions ( $n = 2$ ); compensation reduced systolic MAE by 56.2%.

Core Team Member, Advisor: Prof. Feng Zheng

Mar 2026 – Jun 2026

- **Project: SplatSLAM – Indoor Reconstruction with SLAM and 3D Gaussian Splatting**
  - ▷ Integrated MAST3R-SLAM with Nerfstudio's 3DGS pipeline to generate point clouds, rendered images, and walkthrough videos from monocular video.
  - ▷ Analyzed how training iterations affected PSNR, L1 loss, and visual quality, identifying signs of overfitting and spiky rendering artifacts.

Project Leader, Course Advisor: Prof. Wende Ke

Sep 2025 – Dec 2025

- **Project: ROS 2 Mobile Manipulator for Navigation and Grasping**
  - ▷ Integrated slam\_toolbox, Nav2, and EKF fusion of wheel odometry and IMU data for mapping, localization, and navigation in Gazebo.
  - ▷ Integrated OpenCV visual alignment and arm control for object retrieval, with voice and gesture interfaces using Vosk and MediaPipe.

Project Leader, Course Advisor: Prof. Chenglong Fu

Mar 2025 – Jun 2025

- **Project: Image Processing and Motion Planning for Robotic Drawing**
  - ▷ Developed OpenCV routines for contour extraction and skeletonization to generate drawing paths for a six-axis robotic arm.
  - ▷ Used inverse kinematics in MATLAB to convert planar drawing paths into joint trajectories and verified the resulting motion in simulation.
  - ▷ Implemented tool orientation adjustment along surface normals for simulated drawing on cylindrical surfaces.

Project Leader, Advisors: Prof. Dong Lu, Yuanjian Huang

Dec 2024 – Aug 2025

- **Project: Autonomous Rescue Robot (National Competition)**
  - ▷ Led the development of a mobile robot integrating YOLOv11 object detection and STM32 motion control for autonomous target tracking and retrieval.
  - ▷ Won **National Second Prize** and **Provincial First Prize** in the China Undergraduate Engineering Practice and Innovation Ability Competition.

## SKILLS

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- **Programming:** C/C++, Python, MATLAB
- **Robotics & Simulation:** ROS 2, MuJoCo, Isaac Sim, Isaac Lab, Gazebo, Webots
- **Computer Vision & Visualization:** OpenCV, RViz2
- **Embedded Systems:** STM32, ESP32
- **CAD/CAM & CAE:** SolidWorks, AutoCAD, Edgecam, Abaqus

## HONORS & AWARDS

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- National Second Prize, China Undergraduate Eng. Practice and Innovation Ability Competition 2025
- Provincial First Prize, Guangdong Undergraduate Eng. Practice and Innovation Ability Competition 2025
- First-Class Scholarship for Outstanding Students (Top 3%) 2025
- Second-Class Scholarship for Outstanding Students (Top 8%) 2024, 2026
- Outstanding Student Award 2024, 2025, 2026
- Provincial Second Prize, Guangdong Undergraduate Mechanical Innovational Competition 2025